

STM32F402RE

Drivers

Bare-metal peripheral drivers for the STM32F401RE microcontroller, developed from scratch using reference manual.

Overview

This project provides low-level drivers for the STM32F401RE ARM Cortex-M4 microcontroller. The drivers are written in C and provide a hardware abstraction layer for various peripherals.

Supported Peripherals

Peripheral	Description
GPIO	General Purpose Input/Output
SPI	Serial Peripheral Interface
I2C	Inter-Integrated Circuit
USART	Universal Synchronous/Asynchronous Receiver/Transmitter
RCC	Reset and Clock Control

Project Structure

stm32f401_project/

```
| Drivers/
```

```
| | — inc/                                # Driver header files
```

```
| | | — stm32f401xx.h # MCU specific header
```

```
| | | — stm32f401xx_gpio_driver.h # GPIO driver header
```

```
| |  └─ stm32f401xx_spi_driver.h    # SPI driver header
```

```
└─ stm32f401xx i2c driver.h    # SPI driver header
```

```
| | └─ stm32f401xx_rcc_driver.h    # SPI driver header
```

```
| | | └─ stm32f401xx_usart_driver.h # SPI driver header
```

```
|   └─ src/                                # Driver source files
```

```
└─ stm32f401xx_gpio_driver.c
```

```
└─ stm32f401xx_spi_driver.c
```

```
└─ stm32f401xx_i2c_driver.c
```

```
└─ stm32f401xx rcc driver.c
```

```
|_ stm32f401xx_usart_driver.c
```

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```
└─ Core/
```

$$\vdash \text{inc/}$$

```
|   └─ src/                                # Application source files
```

```
| |— Startup                                # Startup assembly file
```

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```
|— Example Tasks                                # driver applications
```

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[illegible]

Hardware

- MCU: STM32F401RE (ARM Cortex-M4, 84 MHz)
- Board: NUCLEO-F401RE
- Flash: Up to 512 KB
- SRAM: 96 KB

Vedio link: [driver demo videos](#)

0:00 - 0:25 - GPIO LED Toggle
0:25 - 0:51 - USART echo
0:51 - 1:28 - SPI loopback
1:28 - 1:48 - RTC Integration using I2C driver